TRASTUZUMAB

Cardiotoxicity — LVEF monitoring

HER2-positive only

CHEMICAL NAME Recombinant humanised IgG1κ monoclonal antibody against HER2 (ERBB2) receptor

CLASS Monoclonal antibody — HER2-targeted therapy

TRADE NAMES Herceptin · Herceptin SC · Kanjinti · Ontruzant (biosimilars)

THERAPEUTIC USES / INDICATIONS

- HER2-positive breast cancer (early — adjuvant/neoadjuvant for 1 year; metastatic).
- HER2-positive metastatic gastric/gastro-oesophageal junction cancer.

PHARMACODYNAMICS

- Binds **HER2 extracellular domain** → blocks HER2 signalling, ↑ receptor internalisation, induces antibody-dependent cell-mediated cytotoxicity (ADCC).

ADMINISTRATION

- **IV** infusion: loading over 90 min, subsequent over 30 min if tolerated.
- **SC** fixed dose 600 mg over 2–5 min into thigh.
- Check name — trastuzumab vs trastuzumab emtansine (Kadcyla) vs deruxtecan (Enhertu) — different drugs.
- Observe for infusion reactions (6 h after first IV dose, 2 h after later doses per label).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Infusion reactions (chills, fever) — mainly first dose.
- Fatigue, diarrhoea, headache.
- Arthralgia, rash.

ADVERSE EFFECTS (SERIOUS)

- Cardiotoxicity: ↓ LVEF, heart failure (↑ with anthracyclines); usually reversible.
- Pulmonary toxicity (interstitial pneumonitis).
- Oligohydramnios in pregnancy.
- Severe infusion reactions/anaphylaxis.

DOSAGES

- **3-weekly IV:** 8 mg/kg loading, then 6 mg/kg every 3 weeks.
- **Weekly IV:** 4 mg/kg loading, then 2 mg/kg weekly.
- **SC:** 600 mg every 3 weeks (no loading).
- Early breast cancer: total 1 year (17–18 cycles).

CONTRAINDICATIONS & CAUTIONS

- Severe dyspnoea at rest due to advanced malignancy or requiring oxygen.
- Hypersensitivity to trastuzumab or murine proteins.
- Pregnancy (contraception during and 7 months after).

KEY INTERACTIONS

- **Anthracyclines (doxorubicin, epirubicin):** ↑ cardiotoxicity; avoid concurrent use; wait after anthracycline.
- **Paclitaxel:** ↑ trastuzumab levels (minor).

PHARMACOKINETICS

- Catabolised like IgG.
- **t½:** ~28 days (washout ~7 months).

NURSING MANAGEMENT

- Baseline LVEF and every 3 months during treatment.
- Monitor vital signs during infusion; manage reactions (slow/stop, antihistamine, paracetamol).
- Confirm HER2 status and exact drug name.
- Ask about breathlessness, oedema, palpitations.

PATIENT EDUCATION

- Heart scans every 3 months are essential.
- Report breathlessness, ankle swelling, palpitations, cough.
- Use effective contraception for 7 months after the last dose.
- Fever/chills with the first infusion are common.

KEY POINTS

- Anti-HER2 monoclonal antibody.
- Cardiotoxic; LVEF every 3 months.
- 8 mg/kg load, 6 mg/kg 3-weekly (or 600 mg SC).
- Not with concurrent anthracycline; don't confuse with T-DM1.

SOURCES CHECKED **[912]** Herceptin 600 mg Solution for Injection in Vial (UK SmPC) **[913]** Herceptin 150mg Powder for concentrate for solution for infusion (UK SmPC) **[914]** Ontruzant, IV (US label) **[915]** Herzuma, IV (US label)

Full references p. 432–457